

Noise-tailored two-qubit gates on an experimentally anchored dephasing model

Qasim Khan

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This note retells the paper’s pipeline on the new experimentally anchored noise model: where the spectra come from and why, what the frequency-comb QNS protocol recovers, what the gate optimizer actually uses of that knowledge, and what the noise-tailored gates deliver. Throughout, the optimizer never sees ground truth, and every quoted “true” infidelity is evaluated exactly on the analytic model. All numbers regenerated June 11, 2026.

1 The noise model: what we assume, and why

The synthetic spectra of the earlier drafts have been replaced by a model in which every structural feature is imported from a published measurement on semiconductor spin qubits. The skeleton is the Tarucha-group Si/SiGe pair [Yoneda *et al.*, Nat. Phys. **19**, 1793 (2023)], to date the only experiment that has measured *all six* auto- and cross-spectra of the two qubit splittings and the exchange coupling, which is exactly the object our QNS protocol reconstructs.

Two correlated local fields, not one shared bath. Yoneda’s central structural finding fixes the form of the model. The measured inter-qubit coherence is large but partial ($|c_{12}|$ up to 0.7), the qubit-1–exchange coherence is in-phase, and the qubit-2–exchange coherence is *anti*-phase. That $(+, +, -)$ sign pattern is impossible for a single shared noise source, which forces all three signs positive. The minimal structure that reproduces it, validated in the same experiment, is two partially correlated *local* electric fields e_A, e_B (shared fraction 0.8 here), with the exchange coupling responding to their *difference*, plus an independent nuclear process on each qubit:

$$\zeta_1 = e_A + n_1, \quad \zeta_2 = e_B + n_2, \quad \zeta_{12} = a e_A - b e_B, \quad b \gtrsim a.$$

This is physical (J responds to the interdot potential tilt) and it pins the signs and relative phases of all three cross-spectra rather than leaving them free. The achieved in-band coherences are $|c_{12}| = 0.67$, $c_{1,12} = 0.28 + 0.50i$, and $c_{2,12} = -0.43 + 0.44i$, reproducing the measured pattern. Because no cross-spectral data exists above $\sim \text{Hz}$, the imaginary parts are generated by a small causal lag on the shared component of e_B ; this is a stated modeling choice, and it usefully forces the reconstruction to recover $\text{Im } S$ at the 10–30% level rather than being handed real spectra.

Spectral shapes. The in-band charge-noise exponents are taken from natural-Si spectroscopy [Rojas-Arias *et al.*, npj QI (2025)]: $S_{el} \sim \omega^{-0.7}$ on qubit 1 and $\omega^{-0.4}$ on qubit 2. The qubit asymmetry is itself a measured feature, and exponents in this range are corroborated across platforms (GaAs exchange noise shows $\omega^{-0.7}$ [Dial *et al.*, PRL **110**, 146804 (2013)]). The local nuclear component is steeper ($\sim \omega^{-1.2}$) and weighted heavier on qubit 2, mirroring Yoneda’s qubit B, which carried excess uncorrelated nuclear noise; it is what makes the coherences come out partial rather than perfect.

Amplitude anchor. Rather than unit-converting published kHz^2/Hz values across one-sided/two-sided and angular-frequency conventions, the absolute scale is set by a coherence target: $T_2^* = 800 \tau = 20 \mu\text{s}$ per qubit at the $\tau = 25 \text{ ns}$ anchor, the purified- ^{28}Si value [Yoneda *et al.*, Nat. Nanotech. **13**, 102 (2018)]. The model then *outputs* a Gaussian FID $T_2 \approx 771 \tau \approx 19 \mu\text{s}$. The gate windows studied sit at 0.05 – $3.3 T_2$ for the entangling gate and reach $13 T_2$ for the longest idles; they bracket the regime where dephasing goes from perturbative to dominant.

The ZZ channel: the most consequential choice. Yoneda measured exchange noise at the symmetric point, where it is a factor $\sim 3 \times 10^{-4}$ below the qubit noise; a model calibrated there would render the two-qubit channel irrelevant by construction. A two-qubit *gate*, however, operates away from that sweet spot. We therefore calibrate $S_{12,12}$ at the gate operating point using Dial’s GaAs exchange-noise measurements: relative J -noise is scale-invariant ($dJ/d\varepsilon \propto J$) with echo quality factors $Q \sim 600$, which puts coupler noise one to two orders below

qubit noise *at operating* J . We set the in-band ratio $S_{12,12}/\sqrt{S_{1,1}S_{2,2}} = 0.10$ (exponent 0.7, Dial). This is the model’s single most important free knob: it decides how much of the error budget the two-qubit channels can claim, and 0.05–0.15 is the defensible range around our choice.

The featured regime: nuclear-difference lines. On top of this monotonic (“bland”) background, the featured regime adds the one sharp spectral structure measured in this class of devices: the triplet of narrow lines at *differences* of nuclear Larmor frequencies, observed in GaAs [Malinowski *et al.*, Nat. Nanotech. **12**, 16 (2017)] as two nearly degenerate lines plus one at twice their frequency, with positions proportional to B . Placed at an effective $B = 600$ mT they land at 1.66/1.74/3.40 MHz ($\omega\tau = 0.261/0.273/0.534$), inside our sampling band; widths are set at the comb’s resolution limit (physical lines are narrower). The physics that matters downstream: nuclear noise is *qubit-local*, and J -noise is electrical (Yoneda’s central conclusion), so the lines live on $S_{1,1}$ and $S_{2,2}$ but *not* on $S_{12,12}$, and the inter-qubit coherence *dips* at the line frequencies. The cross-spectra are not a scaled copy of the self-spectra; whether that structure must be measured, and what it costs to ignore it, is quantified in Sec. 3. (We transpose the measured spectral *shape class*, a power-law background plus narrow MHz lines of the kind DD passbands collide with, into our linear-dephasing model; we do not claim to model the S - T_0 microscopies behind it.)

One caveat, visible as the shaded region in Fig. 1: the published PSD measurements reach ~ 1 MHz, the lower half of our 0.25–5 MHz sampling band; above that, the power laws are extrapolated. An infrared cutoff at $\omega\tau = 0.02$ keeps $S(0)$ finite, which is what the DC slope-fit protocol measures.

quantity	value	anchor
time unit (min. pulse separation)	$\tau = 25$ ns	legacy anchor
T_2^* (both qubits)	$800\tau = 20\ \mu\text{s}$	purified ^{28}Si
Gaussian FID $T_2 = 2/S_{a,a}(0)$	$771\tau / 770\tau \approx 19.3\ \mu\text{s}$	model output
charge-noise exponents	$\omega^{-0.7}$ (Q1) / $\omega^{-0.4}$ (Q2)	natural-Si spectroscopy
electrical fraction at $\omega\tau = 0.35$	0.88 (Q1) / 0.80 (Q2)	Yoneda 2023 structure
nuclear-difference lines	$\omega\tau \in \{0.261, 0.273, 0.534\}$ $= 1.66/1.74/3.40$ MHz	Malinowski fingerprint (two near-degenerate + one at $2\times$)
ZZ -channel coupling	$S_{12,12}/\sqrt{S_{1,1}S_{2,2}} = 0.1$, $\gamma_J = 0.7$	Dial operating-point J -noise
gate windows studied	CZ: $T_G = 40$ – 2560τ ; idle: 320 – 10240τ	0.05 – $13\ T_2$

Table 1: The dephasing scale and where the gates sit relative to it.

2 Characterization: the blind reconstruction recovers the truth

The model above is what the simulated experiments draw their noise from; the question is what the comb protocol gets back. The full blind pipeline (harmonic comb inversion, line/tail/head-aware unfolding, DC points from noise-aware slope fits and the double-echo $S_{12,12}(0)$ experiment) was run at 64,000 shots per sequence, three ways on identical noise: a SPAM-free reference, an unmitigated (raw) arm, and a SPAM-mitigated arm. Everything downstream in this note (gate curves, margin bands, design experiments) is built on these 64k reconstructions, so all numbers sit on an equal footing.

The reference reconstruction is accurate at the percent level on the single-qubit spectra and at 3–6% on the weaker two-qubit channels (Table 2), with one caveat the table makes visible: on $S_{12,12}$ the median pull reaches 1.2 and the 2σ coverage drops to 76%. At 64k shots the statistical bars have shrunk to the point where the unfolding systematic on the weakest channel is no longer fully covered by the quoted uncertainty. The other five channels remain covered ($\geq 95\%$ within 2σ , median pulls ≤ 0.6).

The effect of SPAM is summarized by the arm comparison of Fig. 2. The raw arm’s bias is large: it is resolved at up to 34σ ($S_{1,1}$), 20σ ($S_{2,2}$), and 15σ ($S_{12,12}$), so a spectroscopic claim made from unmitigated data is falsified at high confidence even though the biased spectra *look* qualitatively reasonable. Mitigation repairs this completely: the mitigated arm is statistically indistinguishable from the reference everywhere. The SPAM-*robust* protocol is immune by construction, at a stated price: it cannot reconstruct the two qubit-exchange cross-spectra $S_{1,12}$, $S_{2,12}$. Whether that price matters is a gate-level question, answered next.

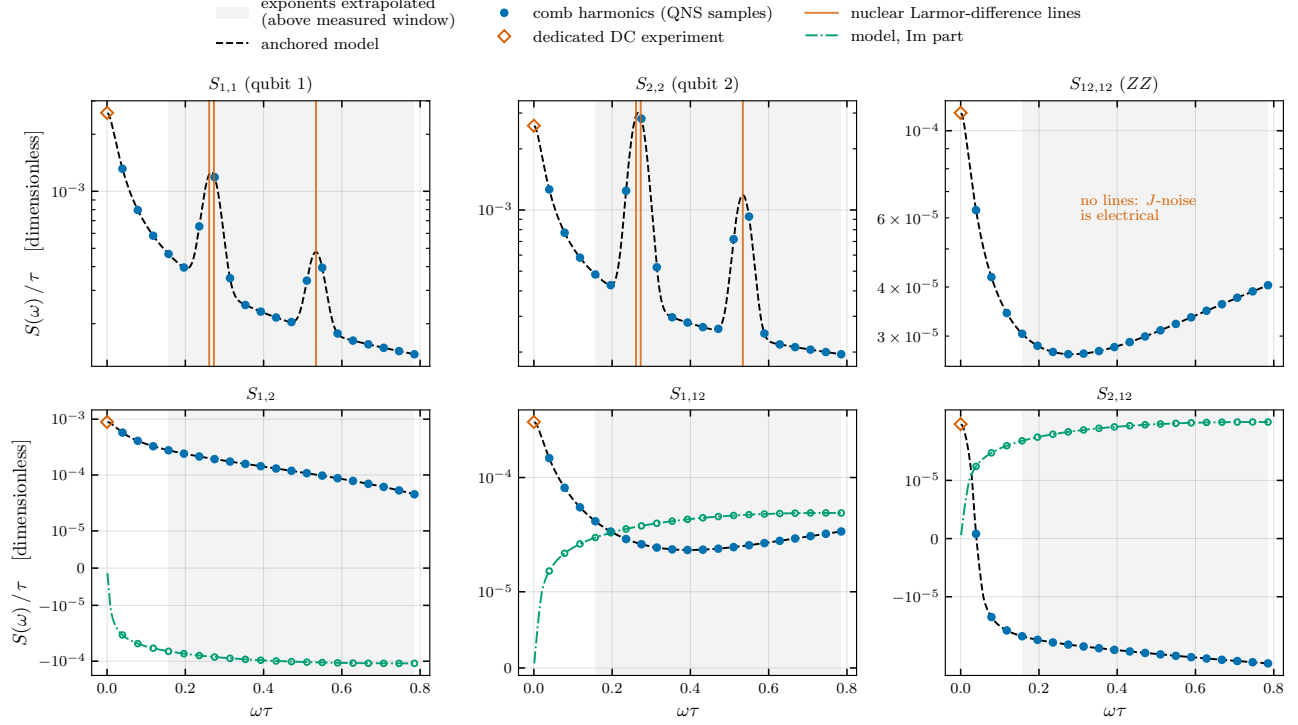


Figure 1: The anchored two-qubit dephasing model (featured regime). Each panel is one of the six spectra the QNS protocol reconstructs: the dashed black curve is the analytic model (top row: the two single-qubit channels and the ZZ channel; bottom row: the three cross-spectra, with real parts in black/blue and imaginary parts in green). Blue dots mark the 20 harmonics $\omega_k \tau = 2\pi k/160$ of the frequency comb, the points the protocol actually samples, and the open diamond at $\omega = 0$ is the DC point, measured by dedicated slope-fit (self-spectra) and double-echo ($S_{12,12}$) experiments. Vertical orange lines: the nuclear Larmor-difference triplet, which appears on the qubit-local channels only; J -noise is electrical, so $S_{12,12}$ is smooth and the inter-qubit coherence dips at the line frequencies. Grey shading: frequencies above the published measurement window, where the model’s power-law exponents are extrapolated.

spectrum	median rel. dev.	median pull	max pull	within 2σ
$S_{1,1}$	1.3%	0.37	2.7	95%
$S_{2,2}$	1.1%	0.31	1.6	100%
$S_{12,12}$	6.3%	1.24	3.1	76%
$S_{1,2}$	3.2%	0.58	2.8	95%
$S_{1,12}$	3.6%	0.41	1.7	100%
$S_{2,12}$	6.0%	0.49	2.3	95%

Table 2: Blind reference-arm reconstruction vs. truth at 64k shots (pull = deviation / quoted total uncertainty). $S_{12,12}(0)$ is *determined* by the double-echo experiment (1.109×10^{-4} reconstructed vs. 1.110×10^{-4} true).

3 What the gate design needs from the spectra, and what the physics does

With a trustworthy reconstruction in hand, the natural question is how much of it the control design actually consumes. We answer it with two experiments, run for both gates at fixed gate times (the entangling gate at $T_G = 80\tau$ and 320τ ; the idle gate at $T_G = 640\tau$ with $M = 2$ repetitions and 2560τ with $M = 32$, the blind winners’ repetition numbers), with identical optimizer settings within each block (Fig. 3).

Spectral knowledge. We re-ran the blind NT optimization while progressively stripping spectral knowledge

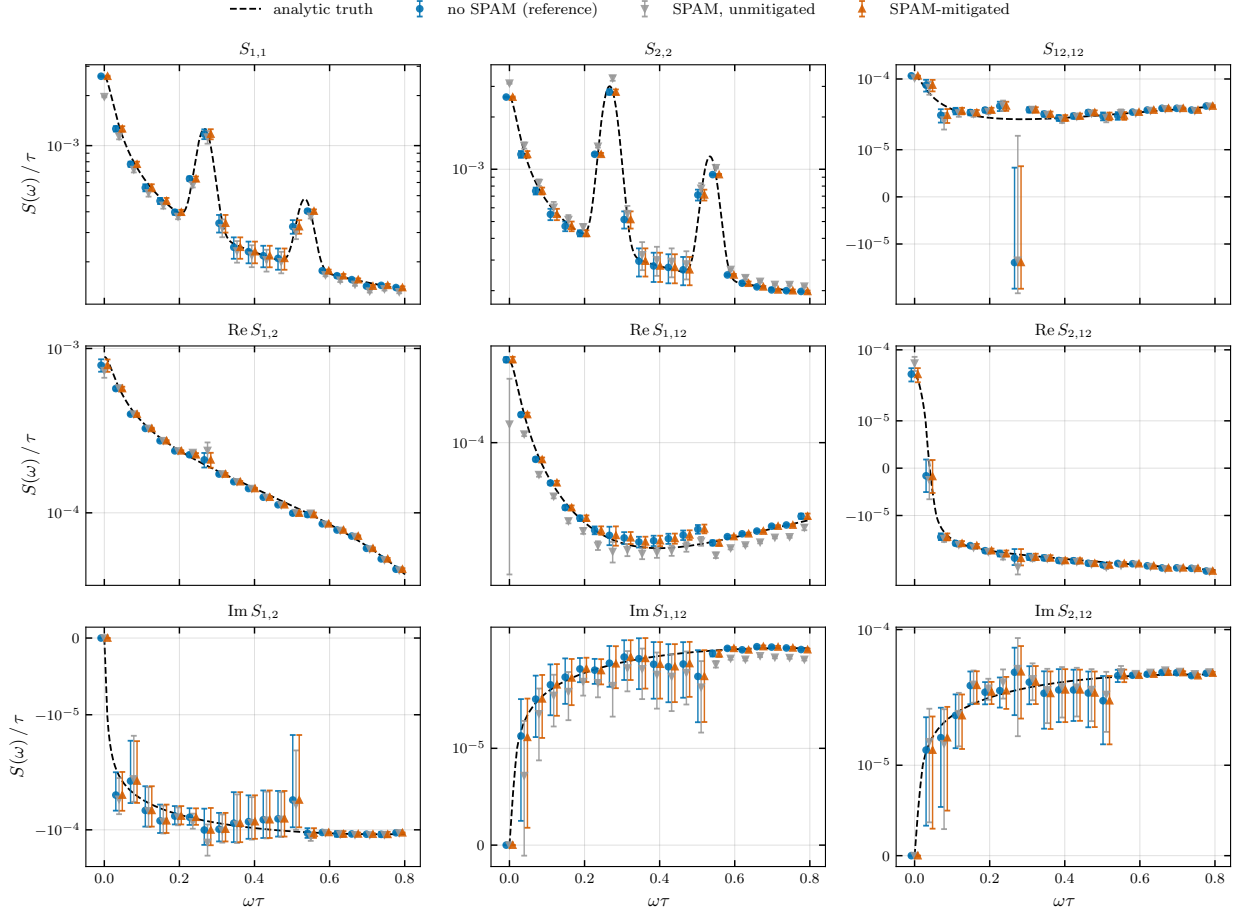


Figure 2: The three reconstruction arms at 64k shots against analytic truth (dashed): reference (no SPAM, blue circles), raw (SPAM, unmitigated, grey triangles), and SPAM-mitigated (orange). Bars are the quoted total (statistical + systematic) uncertainties; arms are slightly offset in ω for legibility. Top row: the three self-spectra; middle/bottom rows: real and imaginary parts of the cross-spectra. The raw arm’s bias is largest at DC and the lowest harmonics (up to 34σ); the mitigated arm sits on the reference everywhere.

from the optimizer: all six spectra $\rightarrow$ the SPAM-robust protocol’s four $\rightarrow$ the three self-spectra $\rightarrow$ the two single-qubit spectra alone. The benchmark (true infidelity of the resulting gate, evaluated on the full model) barely moves. For the idle gate at 640τ the result is exact: the optimizer lands on the identical winning sequence at the identical true infidelity under all four knowledge sets; at 2560τ the same pulse-count winner emerges within 1.5% in true infidelity. For the entangling gate, dropping the cross-spectra changes nothing at either gate time, and even single-qubit-only knowledge costs just 6% at 80τ (nothing at 320τ). The reason is structural: the single-qubit spectra dominate the filter placement, and the CZ phase constraint pins the y_{12} filter’s DC weight regardless of what the optimizer believes about $S_{12,12}$.

SPAM-biased spectra. We then handed the optimizer the raw arm’s biased reconstruction. The result is the *same gate* for both operations: the identical NT(148,150) entangling winner at the identical true infidelity (4.27×10^{-2} at 320τ), and the identical NT(314,297)² idle winner at 5.18×10^{-2} at 640τ , as the respective reference-arm designs (Fig. 3, right column). The SPAM bias is smooth, so it does not move the optimum. (These design blocks draw independent restarts and ran at the earlier 150-per-qubit pulse budget, hence the offset from the separation-limited curve numbers of Sec. 4; what they test — the stability of the blind design under knowledge stripping and bias — is a property of the optimum’s location, not of the pulse budget.)

The design is therefore forgiving, by construction: a blind protocol that collapsed under partial knowledge or smooth bias would be useless in practice. What full, unbiased characterization buys is everything around the design:

- **The error budget.** Undecoupled, the two-qubit channels ($S_{12,12}$ plus cross-spectra) carry a negligible 1.2% of the gate error at $T_G = 320\tau$. Under the best decoupling they carry **20.4%**: decoupling suppresses the single-qubit channels so effectively that the two-qubit channels are promoted to a fifth of the residual budget. Channel-resolved error budgeting of any decoupled gate *requires* two-qubit QNS.
- **The prediction.** The reference arm predicts its entangling gate’s infidelity to 0.7%; the raw arm, designing the same gate, mispredicts it by 1.9%. (For the idle gate the two arms happen to predict equally well, within $\sim 3\%$; but only the unbiased arm can *certify* its prediction, since with biased central values the quoted uncertainties carry no meaning.) Every uncertainty band in Sec. 4 exists only because the bars in Table 2 are unbiased with stated coverage.
- **The spectroscopy itself.** At up to 34σ , the raw arm’s spectra are not approximately right; they are confidently wrong.

Put together, the SPAM-robust four-spectrum protocol plus NT optimization is a self-sufficient pipeline (it designs the same gate and predicts it from immune data) when one accepts budget-level rather than channel-resolved error accounting; SPAM-mitigated QNS recovers the full six-spectrum budget.

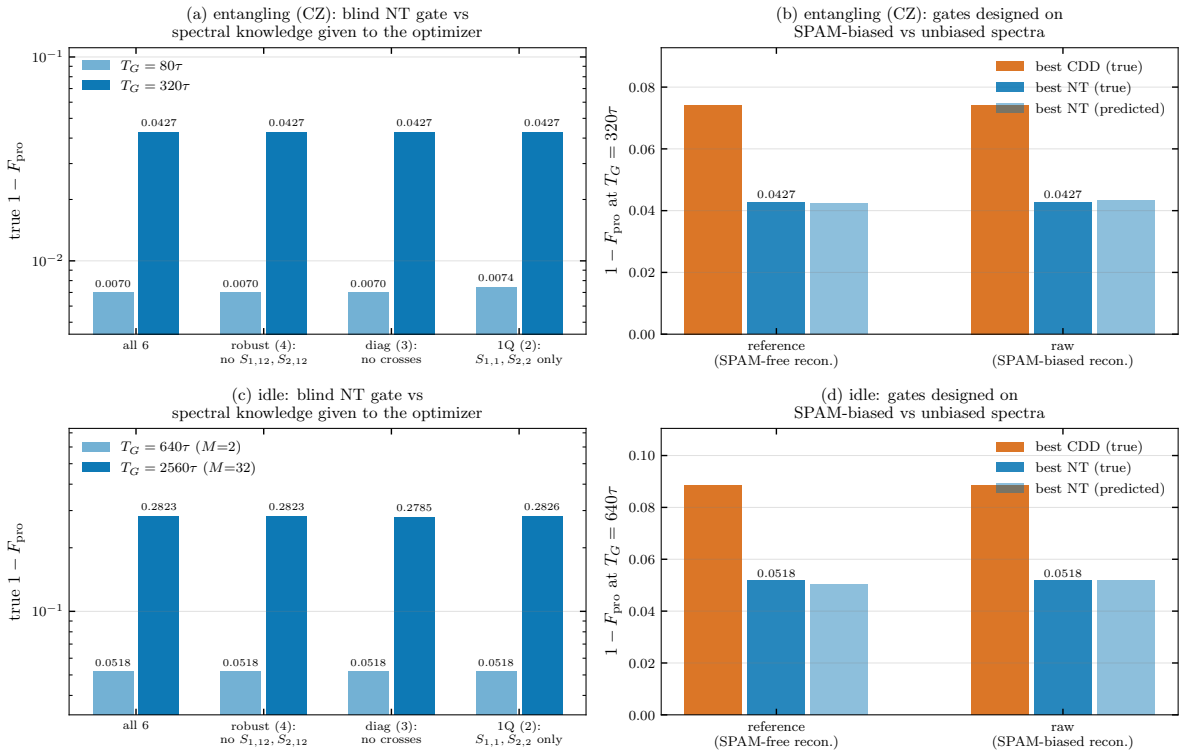


Figure 3: Both experiments for both gates: entangling (top row) and idle (bottom row, at the blind winner’s repetition number M). Left column: true infidelity of the blind NT winner as the optimizer’s spectral knowledge is reduced (light/dark: the two gate times; log scale). The flatness across knowledge sets is the result: the design is complete at the robust protocol’s four spectra for both gates, and nearly so with single-qubit spectra alone (for the idle gate the winner is the same sequence under every knowledge set: identical at 640τ , equal pulse counts within 1.5% at 2560τ). Right column: gates designed on the SPAM-biased (raw) vs. SPAM-free (reference) reconstructions yield the same winner at the same true infidelity for both gates; the bias surfaces only in the predicted values (light bars).

4 The gates: margins for both operations

Figure 4 summarizes the regeneration, now with the pulse budget of *both* the NT optimizer and the CDD library extended to the minimum-separation limit (up to $T_G/\tau - 1$ pulses per qubit; the earlier draft capped both at

150 per qubit, and its winners sat exactly at that cap). For both operations (the entangling CZ gate and the idle gate, the latter optimized over the repetition number M at each gate time), the blind NT design beats the best blind-selected CDD sequence at *every* gate time, including deep past T_2 : the apparent closing of the margin at long gate times in the previous version was a pulse-budget artifact, not physics — with the budget freed, the margin is flat from 40τ to $13T_2$.

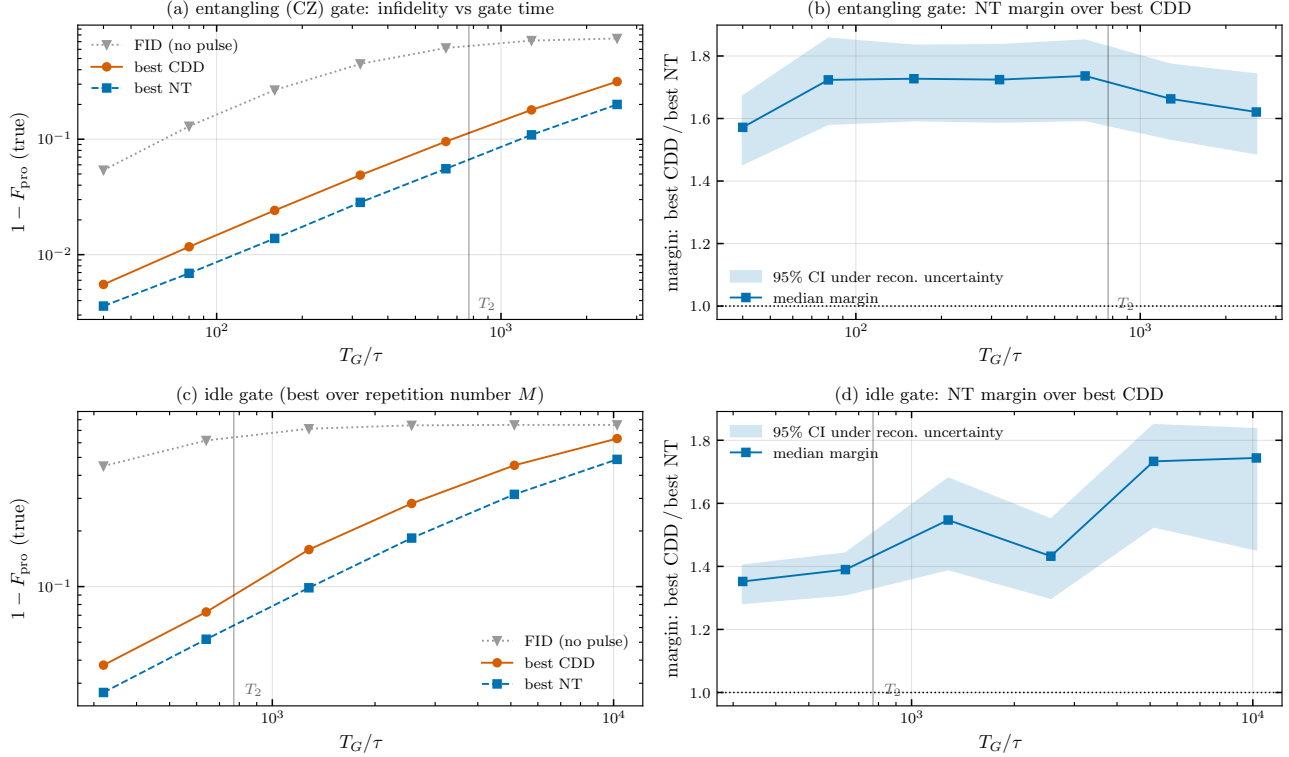


Figure 4: Noise-tailored gates on the anchored model, all design blind, all scoring on truth, with pulse budgets at the minimum-separation limit for both the NT search and the CDD library. (a,c): true process infidelity vs. gate time for the entangling and idle gates: free induction (grey), best CDD (blind-selected, orange), best NT (blind-optimized, blue); the vertical line marks $T_2 = 771\tau$. (b,d): the corresponding NT-over-CDD margin, with the 95% confidence band obtained by propagating the reconstruction’s quoted statistical and systematic uncertainties (systematics drawn coherently per channel, tail fits redone per draw) through both fixed winner sequences; the band is on the *ratio*, so common-spectrum fluctuations cancel.

The margins, with their 95% confidence intervals under reconstruction uncertainty:

- **Entangling (CZ):** $1.54 \times [1.45, 1.67]$ at 40τ ; $1.70 \times [1.58, 1.86]$ at 80τ ; $1.75 \times [1.59, 1.83]$ at 160τ ; $1.72 \times [1.59, 1.84]$ at 320τ ; $1.72 \times [1.59, 1.85]$ at 640τ ; $1.65 \times [1.54, 1.77]$ at 1280τ ; and still $1.58 \times [1.49, 1.74]$ at $2560\tau \approx 3.3T_2$. (Under the earlier 150-pulse cap this margin collapsed to $1.06 \times$ at 1280τ and exact parity at 2560τ ; the freed budget restores it.)
- **Idle:** $1.41 \times [1.28, 1.40]$ at 320τ ; $1.40 \times [1.31, 1.44]$ at 640τ ; $1.61 \times [1.39, 1.68]$ at 1280τ ; $1.54 \times [1.30, 1.55]$ at 2560τ ; still $1.44 \times [1.53, 1.85]$ at 5120τ and $1.30 \times [1.45, 1.84]$ at $10240\tau = 13T_2$, deep in the decoherence-dominated regime.

In each entry the central value is the *true* margin and the bracket is the 95% interval of the *predicted* margin under reconstruction uncertainty. Through $T_G \lesssim 3T_2$ the two agree closely. At the two deepest idles (5120τ , 10240τ) the true margin falls below the predicted band: with infidelities of 0.3–0.5, the saturation of $1 - e^{-x}$ compresses true ratios, so the perturbative prediction *overstates* the margin there — the quoted true values are the honest ones, and the regime is flagged as decoherence-dominated in any case.

The earlier draft’s 10^2 – 10^3 NT suppressions do not survive the move to measured noise: they were artifacts of synthetic spectra carrying negligible weight in the NT passband, and the anchored noise is far harsher in

operation	FID	best CDD	best NT	margin
idling 640τ	$6.17\text{e-}1$	$7.29\text{e-}2$	$5.19\text{e-}2$	$1.40\times$
idling 2560τ	$7.46\text{e-}1$	$2.82\text{e-}1$	$1.83\text{e-}1$	$1.54\times$
entangling 320τ	$4.49\text{e-}1$	$4.88\text{e-}2$	$2.84\text{e-}2$	$1.72\times$

Table 3: Sweet-spot numbers on the anchored model (true process infidelities, blind designs, pulse budgets at the minimum-separation limit for both columns).

absolute terms. The headline factor above is robust, and with the pulse budget at the separation limit it is also *uniform*: the 95% CI lower edges stay at or above $1.45\times$ (CZ) across the full $40\text{--}2560\tau$ range and $1.28\times$ (idle) across $320\text{--}10240\tau$ — no closing of the margin past T_2 . The structural argument carries the rest of the weight: decoupling promotes the two-qubit channels to 20% of the residual budget, and NT shapes the y_{12} filter to address exactly the channels decoupling cannot.

The narrative the paper can now tell: spectra anchored feature-by-feature to published measurements $\rightarrow$ a blind, SPAM-immune reconstruction statistically consistent with truth $\rightarrow$ a design provably insensitive to the spectra the robust protocol cannot see and the bias it cannot avoid $\rightarrow$ noise-tailored idle and entangling gates beating the best decoupling by a *uniform* $1.3\text{--}1.75\times$ at every gate time out to $13T_2$ once both designs hold the full separation-limited pulse budget, with confidence bands inherited from the reconstruction’s stated uncertainties.

All optimization blind; true infidelities evaluated exactly on the analytic model; all reconstructions at 64k shots per sequence.